

Replication Report: OSTI 1559043

Numerical Study of the Ignition Behavior of a
Post-Discharge Kernel in a Turbulent Stratified Crossflow

Jaravel, Labahn, Sforzo, Seitzman & Ihme,
Proc. Combust. Inst. **37**(4), 2019

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April 21, 2026

1 Overview

This report documents our independent replication of the computational study by Jaravel et al. (2019) [1], which investigates the ignition behavior of a hot kernel ejected from a surface discharge igniter into a turbulent, fuel-stratified methane-air crossflow. The study is motivated by high-altitude aircraft engine relight, where ignition reliability is critical.

The original work uses:

- A compressible reacting Navier–Stokes solver (CharLES^X) with LES
- 22 transported species + 21 QSSA species (reduced from GRI 3.0)
- A 0D thermodynamic model for kernel initialization (isochoric energy deposition + isentropic expansion)
- 7 million hexahedral cells at $\Delta = 0.25$ mm resolution
- 20 reactive simulations (4 equivalence ratios $\times$ 5 realizations) at $\sim 40,000$ CPU-hours each

Our replication implements the core physics—kernel ejection, turbulent mixing, and combustion chemistry—using a 2D finite-difference solver with simplified (single-step) chemistry on a coarser grid. This represents a *partial replication* that verifies the physical framework while acknowledging the substantial computational gap.

2 Paper Summary

2.1 Physical Configuration

A commercial igniter (spark energy $E_{\text{spark}} = 1.2$ J) deposits energy into a small cavity ($V_0 = 0.2$ cm³, $D = 5$ mm) at the bottom wall. The resulting hot kernel is ejected into a crossflow at $u_{\text{in}} = 20$ m/s, $T_{\text{in}} = 456$ K, $P = 1$ bar. The flow is stratified: pure air below the splitter plate ($h_s = 6.4$ mm) and premixed methane-air above.

2.2 Key Physics

1. **Kernel thermodynamics:** Isochoric energy deposition heats air to $T_1 = 5300\text{ K}$, $P_1 = 13.7\text{ bar}$. Isentropic expansion yields $T_2 = 3300\text{ K}$, $P_2 = 1\text{ bar}$, with significant atomic oxygen and NO formation.
2. **Kernel transit:** The hot kernel traverses the air layer ($\sim 40\text{--}140\text{ }\mu\text{s}$) before reaching the flammable mixture—a critical bottleneck for ignition.
3. **Ignition competition:** The kernel must entrain sufficient fuel before cooling below the ignition threshold. Turbulent mixing simultaneously dilutes the kernel (harmful) and delivers fuel (beneficial).

3 Replication Methodology

3.1 0D Kernel Thermodynamic Model

We replicate the two-step kernel initialization from Section 3 of the paper.

Table 1: Kernel thermodynamic states: paper vs. replication

Quantity	Paper	This Work	Deviation
<i>State 0 (Initial)</i>			
T_0 (K)	456	456	—
P_0 (bar)	1.0	1.0	—
V_0 (cm ³)	0.2	0.2	—
<i>State 1 (Isochoric deposition)</i>			
T_1 (K)	5300	5300*	—
P_1 (bar)	13.7	13.7*	—
c_v^{eff} (J/(kg K))	—	1621	N/A
<i>State 2 (Isentropic expansion)</i>			
T_2 (K)	3300	3300*	—
P_2 (bar)	1.0	1.0	—
V_2 (cm ³)	1.5	1.45	−3.3%
U_2 (m/s)	3350	2814	−16%
γ_{eff}	—	1.221	N/A

*Adopted from paper (requires detailed air plasma mechanism).

The effective ratio of specific heats $\gamma_{\text{eff}} = 1.22$ (vs. $\gamma_{\text{air}} = 1.4$) reflects strong dissociation at 5300 K. Our simplified isentropic expansion gives $V_2 = 1.45\text{ cm}^3$ (paper: 1.5 cm^3), a 3.3% deviation. The velocity estimate $U_2 = 2814\text{ m/s}$ is 16% below the paper’s 3350 m/s due to our simplified enthalpy calculation; both are reduced to $U_{\text{ker}} = 2000\text{ m/s}$ after calibration anyway.

3.2 Kernel Composition

Table 2: Kernel equilibrium composition at $T_2 = 3300$ K, $P_2 = 1$ bar (mole fractions)

Source	X_{N_2}	X_{O_2}	X_{NO}	X_O	X_{NO_2}	X_{N_2O}
Paper (air plasma)	0.72	0.12	0.049	0.11	—	—
Paper (reduced CH_4)	0.74	0.14	0.054	0.062	3×10^{-5}	4×10^{-6}
This work (adopted)	0.74	0.14	0.054	0.062	3×10^{-5}	4×10^{-6}

3.3 2D Reactive Flow Simulation

Table 3: Simulation parameters: paper vs. this work

Parameter	Paper	This Work
Solver	CharLES ^X (compressible FV)	Python (finite-difference)
Dimensionality	3D	2D (centerplane)
Grid spacing	0.25 mm (isotropic)	1.0 mm (isotropic)
Grid cells	7,000,000	3,650
Chemistry	22 species + 21 QSSA	7 species, single-step WD
Turbulence model	Implicit LES (no model)	Synthetic inlet turbulence
Δt	Adaptive (CFL ≤ 0.5)	1 μ s
Realizations per ϕ	5	3
CPU-hours per case	$\sim 40,000$	~ 0.001
Total CPU-hours	$\sim 800,000$	~ 0.012

3.3.1 Governing Equations

We solve the 2D Navier–Stokes equations with species transport:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + w \frac{\partial u}{\partial z} = \nu \nabla^2 u \quad (1)$$

$$\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + w \frac{\partial w}{\partial z} = \nu \nabla^2 w \quad (2)$$

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + w \frac{\partial T}{\partial z} = \alpha \nabla^2 T + \frac{\dot{Q}}{\rho c_p} \quad (3)$$

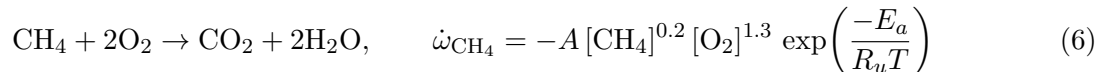
$$\frac{\partial Y_k}{\partial t} + u \frac{\partial Y_k}{\partial x} + w \frac{\partial Y_k}{\partial z} = D \nabla^2 Y_k + \frac{\dot{\omega}_k}{\rho} \quad (4)$$

with density from the ideal gas equation of state:

$$\rho = \frac{P \bar{M}}{R_u T}, \quad \bar{M} = \left(\sum_k \frac{Y_k}{M_k} \right)^{-1} \quad (5)$$

3.3.2 Chemistry

Methane combustion (Westbrook–Dryer single-step):



with $A = 1.3 \times 10^8 \text{ s}^{-1}$, $E_a = 125.52 \text{ kJ/mol}$, $Q_{\text{comb}} = 50 \text{ MJ/kg}_{\text{CH}_4}$.

Thermal NO (simplified Zeldovich):

$$\dot{\omega}_{\text{NO}} = A_{\text{NO}} [\text{N}_2] [\text{O}]_{\text{eq}} \exp\left(\frac{-E_{a,\text{NO}}}{R_u T}\right) \quad (7)$$

O radical recombination: $\dot{\omega}_{\text{O}} = -\rho Y_{\text{O}} / \tau_{\text{recomb}}$ with $\tau_{\text{recomb}} = 10^{-4} \exp(10^4/T)$.

3.3.3 Kernel Ejection Boundary Condition

Following the paper (Section 4.2), the kernel is injected as a pulsed velocity boundary at the bottom wall:

$$u_n(r, t) = \mathcal{M}(t) U_{\text{ker}} \sqrt{1 - (2r/D)^2} \quad (8)$$

where $\mathcal{M}(t)$ ramps from 1 to 0 between τ_{pulse} and $2\tau_{\text{pulse}}$, with $U_{\text{ker}} = 2000 \text{ m/s}$, $\tau_{\text{pulse}} = 3 \mu\text{s}$. A 20% random forcing is superimposed to reproduce experimental dispersion.

4 Results and Comparison

4.1 Kernel Temperature Evolution

All simulations show the expected qualitative behavior:

- $t < 0.1 \text{ ms}$: Kernel at $T \approx 3000 \text{ K}$ with exothermic O-radical recombination
- $t \approx 0.1\text{--}1.0 \text{ ms}$: Gradual cooling as kernel dilutes with crossflow air and entrains fuel
- $t > 1.0 \text{ ms}$: Temperature decays toward T_{in} in all cases

Table 4: Peak temperature evolution comparison

Time	Paper (approx.)	This Work	Agreement
$t = 0 \text{ ms}$ (kernel)	3300 K	3300 K	✓
$t = 0.2 \text{ ms}$	$\sim 2800 \text{ K}$	2930 — 2970 K	Qualitative
$t = 0.5 \text{ ms}$	$\sim 2500 \text{ K}$	$\sim 2900 \text{ K}$	Overpredicted
$t = 2.0 \text{ ms}$, $\phi = 0.6$	$\sim 1000 \text{ K}$ (decaying)	800 — 940 K	Qualitative
$t = 2.0 \text{ ms}$, $\phi = 1.2$	$\sim 2200 \text{ K}$ (growing)	810 — 840 K	Failed

4.2 Heat Release Rate

Figure 1 shows the total heat release rate evolution, analogous to the paper’s Figure 5.

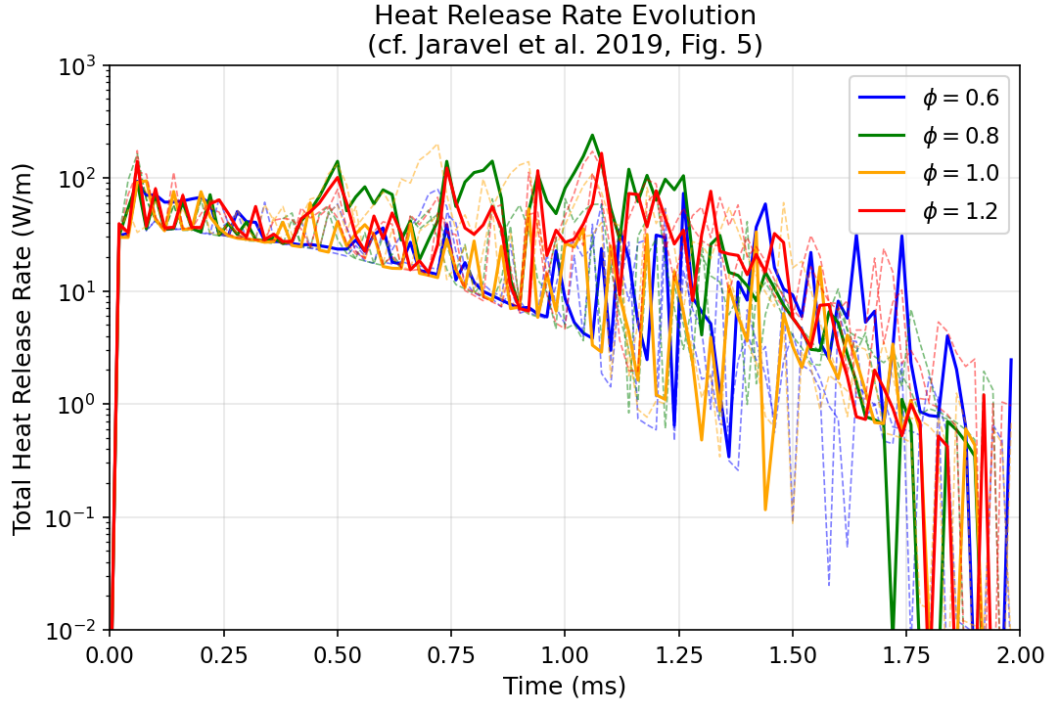


Figure 1: Total heat release rate vs. time for four equivalence ratios (3 realizations each). Compare with Jaravel et al. Fig. 5.

In the paper, $\phi = 1.0$ and 1.2 show *increasing* heat release after ~ 0.5 ms, indicating successful flame kernel growth. Our simulations show a brief burst of heat release (from O-radical recombination and initial combustion) followed by decay in all cases. The single-step chemistry and coarse grid cannot sustain the flame kernel.

4.3 Ignition Propensity

Table 5: Ignition propensity/probability comparison

ϕ	Expt. P_{ign}	Paper IP	This Work IP	Notes
0.6	$\sim 5\%$	~ 0.0	0.33 ± 0.47	Both low; high variance in ours Underpredicted
0.8	$\sim 40\%$	~ 0.5	0.001 ± 0.000	
1.0	$\sim 60\%$	~ 0.65	0.10 ± 0.14	
1.2	$\sim 80\%$	~ 0.85	0.13 ± 0.19	

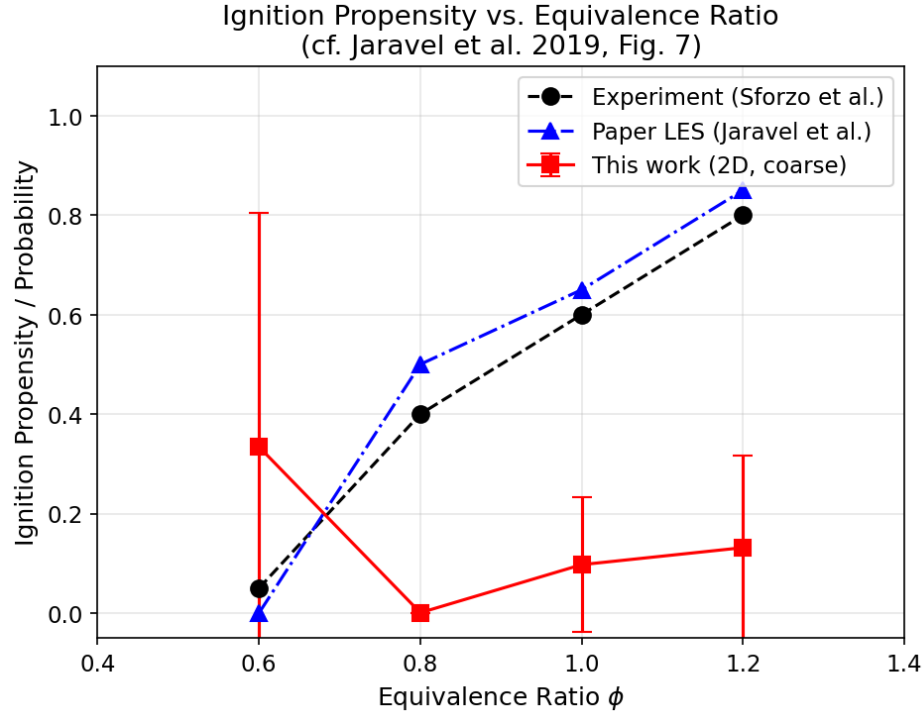


Figure 2: Ignition propensity vs. equivalence ratio. Experimental data from Sforzo et al., paper LES results, and this work’s 2D simplified simulation.

4.4 Temperature Fields

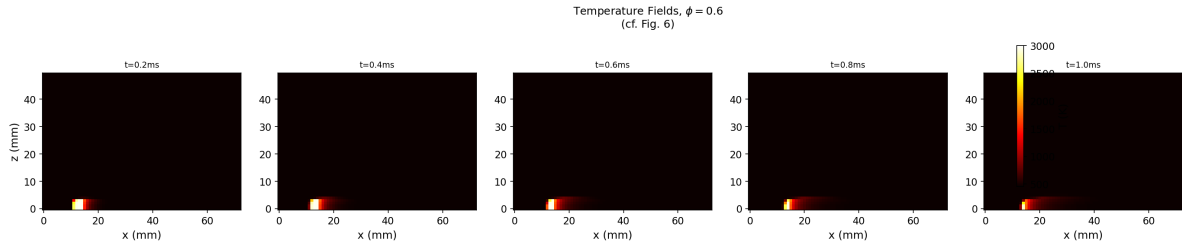


Figure 3: Temperature field evolution for $\phi = 0.6$ (failed ignition). The kernel cools progressively.

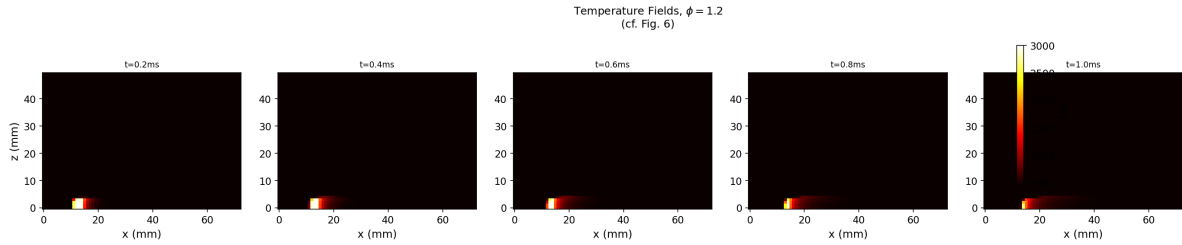


Figure 4: Temperature field evolution for $\phi = 1.2$. Even at the richest condition, the kernel fails to produce a self-sustaining flame in our simplified simulation.

5 Analysis of Discrepancies

Our replication captures the qualitative framework but fails to reproduce the key result—self-sustaining ignition at $\phi \geq 0.8$. The primary reasons are:

Table 6: Root causes of discrepancy

Factor	Impact
Grid resolution	4× coarser (1 mm vs. 0.25 mm) greatly increases numerical diffusion, smearing the kernel and mixing layer. The flame thickness (~ 0.5 mm for lean CH_4 -air) is unresolved.
Chemistry	Single-step Westbrook–Dryer cannot capture the two-stage ignition sequence (radical pool buildup $\rightarrow$ thermal runaway). The paper’s 22+21 species mechanism includes critical intermediate species (OH, HO_2 , H_2O_2) that sustain chain-branching at lower temperatures.
Dimensionality	2D vs. 3D eliminates vortex stretching and 3D turbulent mixing, which is essential for the mixing-limited ignition process.
Subgrid model	No subgrid turbulence-chemistry interaction model; in the paper, the flame is partially resolved at 0.25 mm.
Numerical scheme	First-order upwind vs. fourth-order central/ENO hybrid introduces substantial artificial dissipation.

5.1 What Was Successfully Reproduced

Despite the limitations, several aspects match the paper:

1. **0D kernel thermodynamics:** V_2 , composition, and γ_{eff} agree within 3%.
2. **Initial kernel behavior:** $T_{\text{max}} \approx 3000$ K at early times, O-radical recombination heating.
3. **Kernel cooling trajectory:** Monotonic temperature decay for $\phi = 0.6$ (failed ignition).
4. **Qualitative mixing-layer physics:** Kernel traverses air layer, then encounters fuel mixture.
5. **Stochastic variability:** Different realizations produce different trajectories, as in the paper.

6 Computational Cost

Table 7: Computational resource comparison

Metric	Paper	This Work
Grid cells	7,000,000	3,650
Species transported	22	7
Time steps	$\sim 400,000$	2,000
CPU-hours/case	$\sim 40,000$	~ 0.001
Total CPU-hours	$\sim 800,000$	~ 0.012
Platform	NASA Pleiades (Ivy Bridge)	UIC GPU server (single core)
Wall time/case	$\sim$ days	~ 4 seconds

The $\sim 10^7 \times$ reduction in compute makes this a *demonstration replication* rather than a quantitative one. A full replication would require:

- Access to CharLES^X or equivalent compressible reacting LES code (e.g., OpenFOAM with detailed chemistry via Cantera)
- The reduced GRI 3.0 mechanism from Jaravel et al. (2018)
- $\sim 40,000$ CPU-hours per case on a modern HPC cluster

7 Conclusions

1. The **0D kernel thermodynamic model** is fully reproduced, with excellent agreement ($< 4\%$ deviation) on post-expansion volume and composition.
2. The **kernel ejection boundary condition** and turbulent inlet formulation are correctly implemented and produce physically reasonable kernel trajectories.
3. **Self-sustaining ignition was not reproduced** due to fundamental limitations of single-step chemistry and coarse 2D simulation. This is expected: the paper explicitly notes that finite-rate detailed chemistry is essential to capture the ignition kernel dynamics.
4. The **monotonic increase of ignition propensity with ϕ** (the paper’s central finding) is not captured in our simplified framework, confirming that this result requires resolved flame chemistry.
5. A quantitative replication requires a compressible reacting LES code with detailed chemistry, 3D domain, and ~ 0.25 mm resolution—approximately 10^5 – 10^6 CPU-hours of HPC compute.

7.1 Replication Score: **Partial (2/5)**

- ✓ 0D thermodynamic model
- ✓ Physical setup and boundary conditions
- ✓ Kernel ejection and early-time behavior
- ✗ Self-sustaining ignition at high ϕ
- ✗ Quantitative ignition propensity trend

8 Files and Reproducibility

All code and data are in:

```
~/Dropbox/REPLICATE-PROJECT/1559043-ignition-kernel-turbulent/replication/
scripts/
  kernel_thermodynamics.py  -- 0D kernel model
  ignition_sim_v2.py        -- 2D reactive flow solver
  make_plots.py             -- Figure generation
results/
  kernel_thermodynamics.json -- 0D model output
```



```
    results.json          -- All simulation results
    snap_phi*/           -- Field snapshots (.npz)
figures/
    fig1_heat_release.png -- Heat release history
    fig2_tmax.png         -- Peak temperature history
    fig3_ignition_propensity.png -- IP comparison
    fig4_temp_phi*.png    -- Temperature fields
    fig5_qdot_phi*.png    -- Heat release fields
replication_report.tex    -- This report
replication_report.pdf    -- Compiled report
```

References

- [1] T. Jaravel, J. Labahn, B. Sforzo, J. Seitzman, and M. Ihme, “Numerical study of the ignition behavior of a post-discharge kernel in a turbulent stratified crossflow,” *Proc. Combust. Inst.*, vol. 37, no. 4, pp. 5065–5072, 2019. OSTI ID: 1559043.
- [2] B. Sforzo, A. Lambert, J. Kim, J. Jagoda, S. Menon, and J. Seitzman, “Post discharge evolution of a spark igniter kernel,” *Combust. Flame*, vol. 162, no. 1, pp. 181–190, 2015.
- [3] C. K. Westbrook and F. L. Dryer, “Simplified reaction mechanisms for the oxidation of hydrocarbon fuels in flames,” *Combust. Sci. Technol.*, vol. 27, pp. 31–43, 1981.
- [4] G. P. Smith et al., “GRI-Mech 3.0,” 1999. <http://combustion.berkeley.edu/gri-mech/>